
BBE Sports & Bikes Sales Analytics Dashboard

Consultancy Report & Technical Documentation

Prepared using Power BI Desktop
Data coverage: January 2018 – January 2020
Report version 1.0

This document accompanies the editable Power BI (.pbix) deliverable and should be read alongside it. All figures reference the report as of the version described herein.

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Executive Summary

This report documents the end-to-end development of a sales analytics solution for BBE, a retailer of bikes, parts, accessories, and sports products. The project covered data cleaning and preparation in Power Query, a star-schema data model, a library of DAX measures, and a three-page executive dashboard (Executive Overview, Product Performance, and Customer & Geography).

The analysis confirms that **Bikes account for approximately 95% of total revenue** (\$21.2M of \$22.2M), with sales concentrated in a small number of high-performing SKUs — the top six best-selling products alone belong to a single model family (Mountain-200). Customer acquisition grew substantially through 2019, and a clear positive relationship was found between customer income and average spend. Several data quality issues were identified and documented during the cleaning process (Section 2), and all reported figures should be interpreted alongside the limitations in Section 3, particularly the incomplete January 2020 data window.

Total Sales	\$22.20M
Gross Margin	41.46%
Total Orders	25,410
Total Customers	18,220
Products Sold	133 of 606 catalogued
Countries Covered	6

1 Methodology

The project followed a structured analytics workflow, moving through seven sequential phases:

Phase	Description	Key Deliverable
1. Business Understanding	Reviewed client brief; identified product, geography, timing, and customer as core analytical dimensions	Scope confirmation
2. Data Understanding	Column-by-column profiling of Products, Customers, Orders, Promotion	Data profiling notes
3. Data Cleaning (Power Query)	Type casting, decoding coded fields, disguised-null handling, duplicate removal	4 cleaned queries + Products_Working
4. Data Modeling	Star schema construction, relationship configuration, Date table	Power BI data model
5. DAX Measures	~25 measures across KPIs, time intelligence, ranking, segmentation	Measures library
6. Dashboard Design	3-page report under one design system, with navigation	Interactive .pbix report
7. Insights & Recommendations	Synthesis of findings into business recommendations	This report, Section 4

Business Understanding — Reviewed the client brief and identified the core business context: BBE sells bikes, parts, accessories, and sports apparel, and required a dashboard covering product, geographic, temporal, and customer performance.

Data Understanding — Profiled all four source tables (Products, Customers, Orders, Promotion) column by column, documenting data types, null patterns, and referential integrity before any cleaning was applied.

Data Cleaning (Power Query) — Applied documented, auditable transformation steps to each table: explicit type casting, decoding of coded fields (ProductLine, Class), replacement of disguised missing values, and removal of exact duplicate rows. A dedicated **Products_Working** table was derived to exclude non-sellable manufacturing components from the analytical model.

Data Modeling — Built a star schema with **Orders** as the central fact table, related to three dimension tables (Products_Working, Customers, Promotion) and a custom Date table generated in Power Query and marked as the model's official date table to support time-intelligence calculations.

DAX Measures — Developed a library of approximately 25 measures spanning core KPIs, time-intelligence (YoY, YTD, MTD, rolling averages), product ranking, customer segmentation (new vs. returning), and geographic contribution.

Dashboard Design — Delivered a three-page report (Executive Overview, Product Performance, Customer & Geography) under a single design system — a consistent color palette, typography, and layout template — with cross-page navigation.

Insights & Recommendations — Synthesized findings from every phase into the business insights presented in Section 4.

2 Data Cleaning Steps & Key Assumptions

2.1 Products Table

Finding	Decision
209 rows had no ProductCategoryName/ProductSubcategory	Identified as internal manufacturing components (not sellable end-products); excluded from the analytical Products_Working table but retained in the original Products query for reference
Color and SizeRange contained the literal text "NA" rather than true blanks (254 and 307 rows) – masked by default spreadsheet-reading behavior during initial profiling	Explicitly replaced with true nulls in Power Query to prevent "NA" appearing as a false category in visuals
ProductLine and Class were single-letter coded fields (R/M/T/S, H/M/L)	Decoded to full descriptive values based on the known AdventureWorks source-schema convention – flagged as an external assumption, not confirmed by the data itself
Size and SizeRange mixed numeric (cm) and text (S/M/L/XL) values in the same column	Standardized to text type to prevent load errors; no value alteration
16 instances of two distinct ProductKey values sharing an identical ProductName	Documented; ranking/grouping by name may combine these as a single commercial item

2.2 Customers Table

Finding	Decision
Title column 99.5% blank	Removed – no analytical value
1,431 customers (7.7%) had NumberChildrenAtHome greater than TotalChildren – a logical impossibility	Not corrected (no reliable basis for a fix); flagged with a DataQualityFlag_Children column for transparency
HouseOwnerFlag stored as Y/N text	Converted to a true logical (Boolean) column IsHouseOwner
All customer emails share the domain adventure-works.com	Confirms the dataset originates from Microsoft's AdventureWorks sample database; demographic patterns should be read as illustrative rather than representative of a real customer base

2.3 Orders Table (Fact Table)

Finding	Decision
24 fully duplicate rows	Removed via Table.Distinct
5 rows (all dated 27-Jan-2020) had a blank ProductKey	Retained (real sales value exists) but flagged via a Boolean HasProductLink column; excluded from product-level breakdowns through relationship propagation
No Quantity column exists; SalesAmount equals UnitPrice in every row	Core assumption: each order line represents exactly one unit sold
ShipDate is always exactly OrderDate + 7 days; DueDate is always exactly OrderDate + 12 days, with zero variance across all 58,178 rows	Indicates these fields are synthetically derived rather than operationally variable; documented as a data-realism limitation
ProductStandardCost in Orders matches Products[StandardCost] exactly in every row	Retained deliberately (standard fact-table practice: cost-at-time-of-sale), though currently identical to the dimension value
2020 data covers only 1–27 January (partial month)	All time-based visuals display an explicit "partial data" callout; year-over-year comparisons involving 2020 should be read with caution

2.4 Promotion Table

Finding	Decision
Only 4 of 16 registered promotions were ever applied to an order	Documented; most promotional programs in the reference table saw no recorded usage
Discount percentages (e.g., 2% on Promotion #2) are not reflected in the actual UnitPrice charged	Critical limitation: promotion impact on realized revenue cannot be quantified from this dataset; promotions should be treated as a categorical label only

2.5 Cross-Cutting Assumption

Because Orders covers only January 2018 – January 2020, it does **not** represent a customer's complete purchase history. The Customers[DateFirstPurchase] field – not the earliest OrderDate found in Orders – was used as the authoritative source for identifying new vs. returning customers.

3 Data Limitations

The following limitations should be considered when interpreting any figure in this dashboard:

- 1. Incomplete final period.** The dataset ends on 27 January 2020. Any month-, quarter-, or year-level total involving January 2020 reflects 27 days of activity, not a full period. Comparisons against earlier full periods will understate 2020 performance and should not be read as a decline in the business.
- 2. Promotion impact cannot be measured.** Because discount percentages are not reflected in the realized UnitPrice, this dashboard cannot answer "how much extra revenue did a given promotion generate." Promotion fields should be treated as descriptive labels only, not as a revenue driver.
- 3. Unit-level assumption.** With no Quantity field in the source data, every order line is assumed to represent one unit sold. If the true source system allows multi-unit lines, all "units sold" and per-unit metrics in this report would understate true volume.
- 4. Synthetic date behavior.** Shipping and due dates follow fixed, invariant offsets from the order date (+7 and +12 days respectively) across the entire dataset. Any analysis of fulfillment lead time would not reflect real operational variability, because none exists in this sample.
- 5. Sample data source.** All customer records originate from Microsoft's AdventureWorks reference dataset (identifiable via the shared adventure-works.com email domain). Demographic and behavioral patterns should be treated as illustrative of the methodology, not as a finding about BBE's actual customer base, until validated against BBE's real records.
- 6. Partial customer history.** The Orders table spans only January 2018 – January 2020. Customers who made their first purchase before 2018 are correctly identified as returning customers only because the model relies on Customers[DateFirstPurchase], an external field – not because their full order history is present in this dataset. Any customer-level lifetime metric would be incomplete.
- 7. Duplicate product naming.** Sixteen distinct ProductKey values share an identical ProductName with the same color and size. Any visual or ranking grouped by product name (rather than key) treats these as a single item, which may slightly overstate or understate individual SKU performance.
- 8. Five unlinked order lines.** Five order lines (0.01% of all orders, all dated 27 January 2020) carry no valid product reference and are therefore excluded from every product- and category-level breakdown, while remaining included in order- and revenue-level totals.

4 Key Business Insights

Each insight below follows a consistent structure: **what happened**, the **evidence** behind it, **why it matters** commercially, and the resulting **management implication**.

Insight 1: Revenue is heavily concentrated in a single category

What happened: Bikes generate approximately 95% of total revenue (\$21.2M of \$22.2M), with Accessories and Clothing together contributing less than \$1M combined.

Evidence: Category breakdown (Executive Overview, Product Performance pages).

Why it matters: A single-category dependency exposes the business to concentrated risk – any disruption to bike supply, demand, or pricing has an outsized effect on total revenue.

Management implication: Evaluate whether Accessories and Clothing are being actively cross-sold with bike purchases, and assess whether their small contribution reflects a genuine market opportunity or a deliberate strategic focus on bikes.

Insight 2: Best-selling products are concentrated in two model families

What happened: The top six best-selling individual products are all Mountain-200 variants (different colors/sizes); the next four are all Road-250 variants.

Evidence: Top Products ranking (Executive Overview).

Why it matters: Commercial success is driven by two core models, not by broad product-line diversity. Inventory, marketing, and manufacturing capacity are effectively riding on a small number of designs.

Management implication: Consider expanding color/size variants of Mountain-200 and Road-250 rather than investing in entirely new models, and stress-test supply continuity for these two lines specifically.

Insight 3: The gap between top and bottom performers is extreme

What happened: The best-selling product generated \$1.37M in revenue; the ten lowest-performing sold products generated between \$2.4K and \$12.0K each – a difference of roughly two orders of magnitude.

Evidence: Top Products vs. Bottom Products charts (Product Performance page).

Why it matters: Products at the bottom of the range may not justify their shelf space, marketing attention, or inventory carrying cost.

Management implication: Review the commercial case for continuing to stock the lowest-performing accessory items (e.g., Racing Socks, Patch Kits), or consider bundling them with high-performing bikes rather than selling as standalone SKUs.

Insight 4: Customer acquisition accelerated sharply in 2019

What happened: New customer acquisition grew from roughly 3,200 in 2018 to over 12,500 in 2019 – nearly a fourfold increase – before naturally tapering in the partial 2020 window.

Evidence: New vs. Returning Customers chart (Customer & Geography page).

Why it matters: 2019 represents the business's strongest growth period on record in this dataset; understanding what drove it (marketing campaigns, new geographies, promotions) is valuable for replication.

Management implication: Investigate what changed operationally or in marketing spend between 2018 and 2019, and assess whether the same drivers can be reactivated.

Insight 5: Higher-income customers spend more, consistently

What happened: A clear positive relationship exists between customer yearly income and average spend per customer, visible across the full income range.

Evidence: Income vs. Average Spend scatter chart (Customer & Geography page).

Why it matters: Income is a usable segmentation variable for targeting and personalization.

Management implication: Consider income-tiered marketing (e.g., premium bike lines promoted to higher-income segments) once this pattern is validated against BBE's real customer data (see Limitation 5).

Insight 6: Sales are concentrated in two countries

What happened: The United States and Australia together account for roughly 60% of total revenue (\$6.9M and \$6.5M respectively), compared to \$1.4M–\$2.8M for each of the remaining four countries.

Evidence: Sales by Country chart (Executive Overview page).

Why it matters: Growth in the four secondary markets (UK, Germany, France, Canada) represents a comparatively untapped opportunity relative to the two lead markets.

Management implication: Evaluate whether the gap reflects market size/saturation or under-investment in secondary markets, and consider a targeted growth initiative in the strongest of the four (United Kingdom, at \$2.8M).

Insight 7: Promotion programs show negligible utilization and unmeasurable impact

What happened: Only 4 of the 16 registered promotions were ever applied to a real order, and the discount percentage on those promotions is not reflected in the price actually charged.

Evidence: Promotion table analysis (Section 2.4 / Limitation 2).

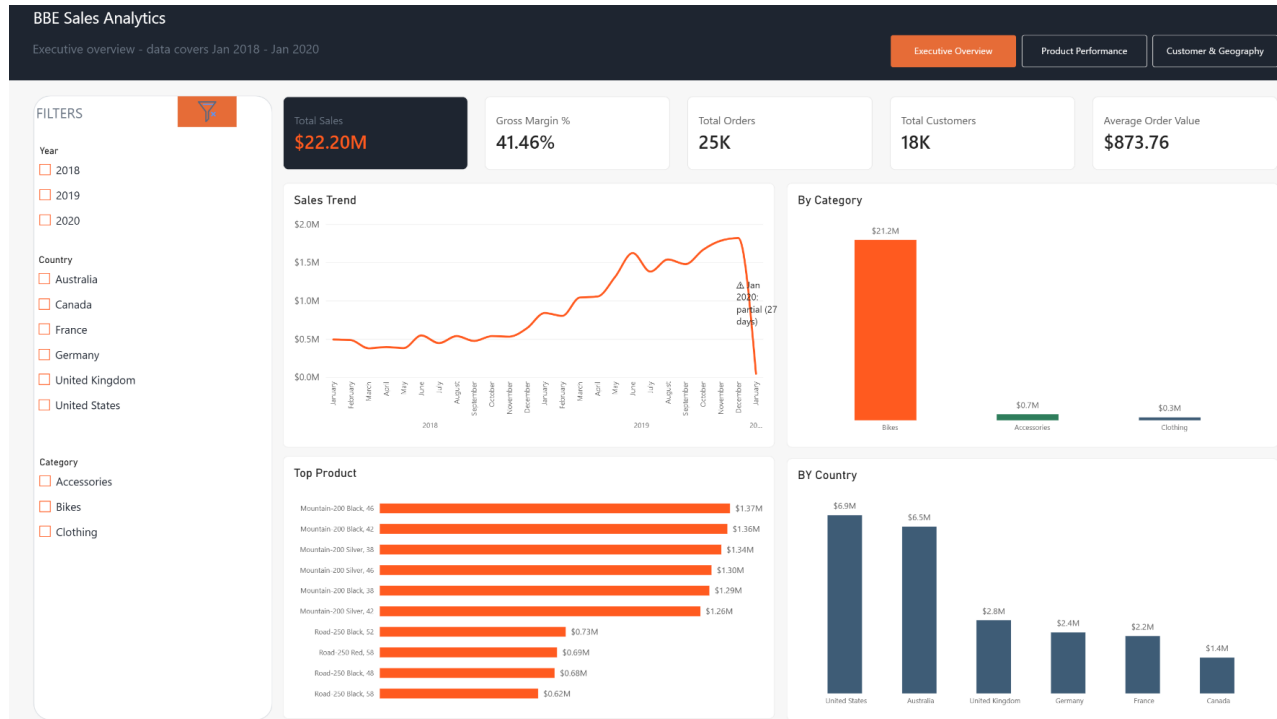
Why it matters: The business cannot currently determine whether any promotion has driven incremental sales, meaning promotional spend and design decisions are effectively unguided by data.

Management implication: Implement discount and quantity tracking at the point of sale so that future promotional campaigns can be measured for incremental revenue, not just recorded as a label.

5 Dashboard Screenshots

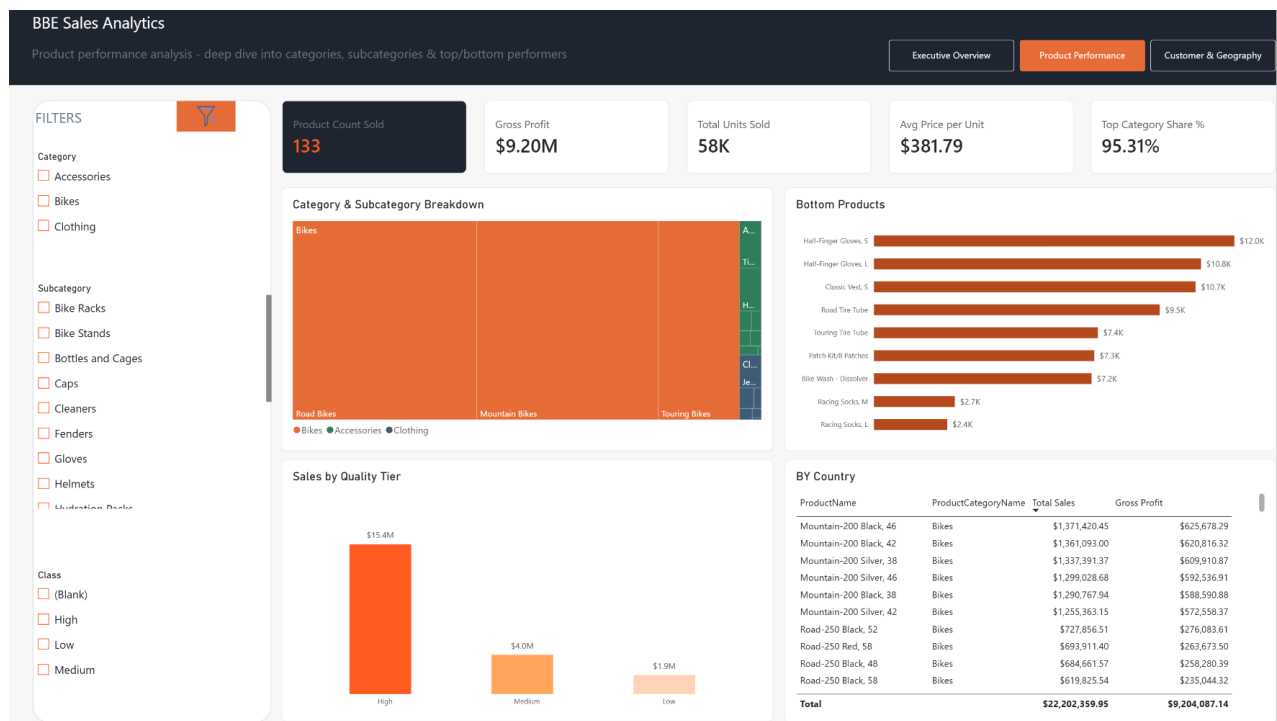
The dashboard consists of three pages sharing a single design system (color palette, typography, and layout template), with cross-page navigation buttons at the top of each screen.

5.1 Executive Overview



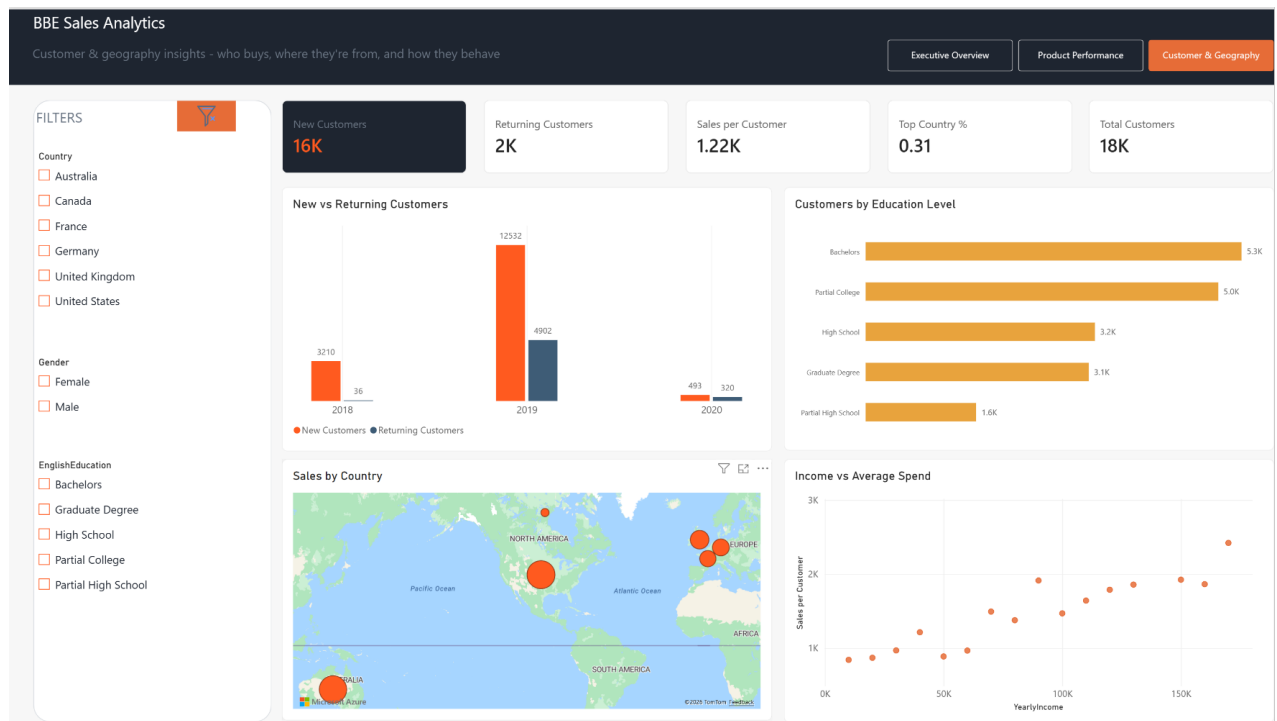
High-level KPIs, sales trend, category split, top products, and country performance.

5.2 Product Performance



Category/subcategory breakdown, top and bottom performing products, quality-tier sales, and full product detail table.

5.3 Customer & Geography



New vs. returning customers, education-level distribution, geographic sales map, and income-vs-spend relationship.

6 Technical Documentation & DAX Reference

6.1 Data Model

The model follows a star schema. **Orders** is the central fact table, related to three dimension tables via single-direction, one-to-many relationships: **Products Working** (on ProductKey), **Customers** (on CustomerKey), and **Promotion** (on PromotionKey). A dedicated **Date** table, generated in Power Query and marked as the model's official date table, relates to Orders on OrderDate only – DueDate and ShipDate were deliberately left unrelated, as both are fixed, non-variable offsets of OrderDate and add no independent analytical value (see Section 2.3).

6.2 Core Measures

Measure	Logic (simplified)
Total Sales	SUM(Orders[SalesAmount])
Total Orders	DISTINCTCOUNT(Orders[SalesOrderNumber])
Gross Profit	[Total Sales] - [Total Cost]
Gross Margin %	DIVIDE([Gross Profit], [Total Sales], 0)
Sales YoY %	DIVIDE([Total Sales] - [Sales PY], [Sales PY])
Product Rank	RANKX(ALL(Products_Working[ProductName]), [Total Sales], , DESC, Dense)
New Customers	Orders where OrderDate = Customers[DateFirstPurchase] (via RELATED)
% Orders without Product Link	Share of order lines with no valid ProductKey (COALESCE-guarded DIVIDE)

6.3 Notable DAX Patterns Used

- **Time intelligence** (Sales PY, Sales YTD/MTD, Rolling 3-Month) relies on the Date table being marked as the model's official date table.
- **Ranking** (Product Rank) filters on ALL(Products_Working[ProductName]) so that a single visual filter on product name does not distort the ranking universe.
- **New vs. returning segmentation** uses the Customers table's own DateFirstPurchase field rather than recalculating it from Orders, because Orders only covers a partial window of each customer's true history (Limitation 6).
- **Guarded division**: every ratio measure uses DIVIDE with an explicit zero fallback, and COALESCE is used where the numerator itself can return a true blank rather than a zero, to avoid "—" placeholders appearing under valid filter contexts.

End of report. Refer to the accompanying .pbix file for the live, interactive model, measures, and Power Query steps referenced throughout this document.